Swetha Ashok

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Education

University of California, Berkeley **May 2028**
B.S. Electrical Engineering and Computer Science, Minor in Energy Engineering **GPA: 3.5/4.0**
Relevant Coursework: Circuits & Devices, Java and Data Structures, PCB Design, C and Computer Architecture, Microfabrication Technology, Microelectronic Devices, Thermodynamics, Digital Design & FPGAs (Spring 2027), PV Cells (Spring 2027), Introduction to Power Electronics (Spring 2027)

Skills

Hardware: PCB design and bring-up, ESP32, embedded systems, soldering, debugging, oscilloscopes, digital multimeters, parameter analyzer
Software: C, Python, Java, RISC-V, Github, MS Suite, LaTeX
CAD: SOLIDWORKS, Onshape, Autodesk Fusion 360, KiCAD, LTSpice

Technical Experience and Projects

Cleanroom Lab: Microfabrication Technology Course **Fall 2026**
- Fabricating IC on silicon wafer in a cleanroom environment, executing full-cycle planar processing steps from blank substrate to finished device with 4 layers (Active Area, Poly-Si, contact holes, and Al interconnects).
- Conducting characterization and parameter verification to evaluate device functionality and yield across process steps.

CalSol - Solar Car Racing Team **Aug. 2024 - PRESENT**
Electrical Engineer, Solar Subteam Advisor *Aug. 2026 - PRESENT*
- Designed a CAN interface PCB in team of 3 to translate communications between 500 kbps MPPT controllers and the 1 Mbps main vehicle bus while stepping down bus voltage from 24V to 12V for system electronics.
- Spearheaded design and hardware build of a mini solar car project, using repurposed PV cells and material.

Solar Project Manager *Aug. 2025 - May 2026*
- Led a 20+ member team in training for solar module wiring, bypass diode soldering, and PV cell integration of array into battery via MPPT, coordinating with electrical and mechanical requirements for expedited timelines.
- Pivoted subteam strategy after timeline shifts through mock integrations, evaluating in-house PV cell encapsulation, benchmarking peer solar architectures, and establishing a broader multi-generation roadmap.

Project: PCB Retro Synth, *Co-Designer and Developer* **Spring 2026**
- Delivered a fully functional 12-note analog PCB synthesizer on a 12-week development timeline by collaborating in a 3-person engineering team across design, hardware bring-up and custom firmware.
- Designing schematics and layout with KiCAD, managing BOM to meet performance and budget constraints.

Leadership and Extracurricular Activities

Executive, Treasurer, Society of Women Engineers **Aug. 2024 - PRESENT**
- Accelerated reimbursement turnarounds by 75% across 100+ active members, managing budget allocations and financial counseling for 30+ distinct programs.
- Organizing logistics and preparing 20 attendees for industry recruitment through career-prep workshops for WE Annual and Regional Conferences.

Peer Adviser, UC Berkeley College of Engineering, Student Services **Aug. 2026 - PRESENT**
- Serve as the sole hardware-focused EECS peer adviser, delivering specialized guidance on degree planning, course selection, hardware tracks, campus resources, and career pathways.
- Advise prospective and undergraduate engineering students on college requirements, major options, and administrative policies such as course planning and major changes, serving over 100 students to date.

Executive, Secretary, Institute of Electrical and Electronics Engineers **Aug. 2026 - PRESENT**
- Manages records, supports officer initiatives, and guides strategic decision-making by the Executive Board.
- Bolster new outreach initiatives focused on increasing engagement with community colleges and women in electrical engineering, expanding member participation across chapter activities.